

Objective Functions as Governance Choices: An Agent-Based Exploration of Efficiency and Equity in Humanitarian AI

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Abstract. Artificial intelligence (AI) and automated decision-support systems are increasingly used to allocate scarce resources in humanitarian and disaster response. Governance debates around these systems have focused on fairness, transparency, accountability, explainability, contestability, and bias. Less attention has been paid to an upstream design choice about the objective function, or what a system is asked to optimize. We argue that this choice is itself a governance decision, and that its distributional consequences are rarely tested directly. To examine this claim, we develop a simple agent-based model of humanitarian aid distribution, in which households with heterogeneous needs and vulnerability receive aid from agencies operating under scarcity. Holding the population, resources, and operating conditions constant, we vary only the allocation objective, comparing throughput-based allocation, which prioritizes proximity and efficient delivery, with vulnerability-based allocation, which prioritizes households identified as most in need. Changing the objective changes who is reached and how the unmet need is assisted. Throughput-based allocation reduces average need across the population and reaches substantially more households, while vulnerability-based allocation concentrates assistance among a smaller, higher-need group. The choice between throughput and vulnerability matters more than the specific way vulnerability is measured. We conclude that objective-function selection should be treated as a governance choice, made explicit, contestable, and open to audit, rather than as a neutral technical parameter.

Keywords: Humanitarian AI, Objective functions, Agent-based modeling, AI governance, Resource allocation, Efficiency-equity tradeoff.

1 Introduction

Artificial intelligence (AI) and automated decision-support systems are increasingly used across humanitarian and crisis response. These applications support forecasting [1], risk mapping [2], damage assessment [3, 4], logistics planning [5], biometric registration [6], and aid allocation [7, 8]. In humanitarian response, these tools are often

valued because they can process large amounts of data quickly and support decisions under uncertainty, incomplete information, and severe resource constraints [9]. For humanitarian agencies, the appeal is practical because AI promises faster assessment, wider reach, and more efficient allocation of limited resources [2, 7, 10, 11].

As these tools become more embedded in humanitarian decision-making, governance debates have focused on fairness, bias, transparency, accountability, explainability, contestability, and human oversight [11, 12]. However, AI systems may reproduce historical inequalities, make affected populations less visible, or make it harder to identify who is responsible for a decision [13-15]. In humanitarian settings, these risks are especially serious because affected communities may have a limited ability to understand, challenge, or refuse automated forms of assessment or processing [2, 7, 10, 16].

Yet fairness and accountability are not only downstream questions about whether a system produces biased outputs or whether a decision can be explained after deployment. There are also upstream questions about how a system defines “success.” Decisions about what to optimize, what to measure, and which tradeoffs to accept shape what the system can see and what it treats as relevant [11]. A tool optimized for speed, coverage, predictive accuracy, or logistical efficiency may produce different outcomes than one optimized for vulnerability reduction or equitable distribution. These choices may appear technical, but they also encode institutional priorities and normative judgments [11, 17-19].

This paper examines objective-function selection as a governance choice in humanitarian aid allocation. We ask, *How do different allocation objectives shape distributional outcomes in humanitarian aid systems?* To explore this question, we develop a simple agent-based model of humanitarian aid distribution. The model compares throughput-based allocation, which prioritizes proximity and efficient delivery, with vulnerability-based allocation, which prioritizes households identified as most vulnerable. The model does not reproduce a deployed humanitarian AI system. Rather, it provides a simplified environment for isolating how different allocation objectives shape distributional outcomes under scarcity. The results show that changing the allocation objective alters both average unmet need and the number of households reached. We therefore argue that objective functions should be treated not as neutral technical parameters, but as governance choices with distributive consequences [11, 18-20].

2 Governance, Optimization, and Humanitarian AI

AI governance scholarship has developed a strong vocabulary around fairness, transparency, accountability, explainability, contestability, and human oversight. These concerns are especially important in humanitarian and disaster settings, where decisions are made under urgency, uncertainty, and resource scarcity. AI systems used for forecasting, registration, triage, and aid allocation may reproduce historical inequalities, rely on incomplete data, or make it difficult for affected populations to understand and contest decisions. For example, the Population Registration and Identity

Management Eco-System (PRIMES), an infrastructure used by the UNHCR to manage aid eligibility across countries, illustrates the high-stakes nature of optimization objectives. The system is mathematically optimized for “identity uniqueness,” with success primarily measured by the breadth of biometric enrollment coverage [11, 21]. Humanitarian AI governance for use cases like these for displaced persons often focuses on preventing bias, improving transparency, preserving human control, and clarifying responsibility across agencies, developers, donors, and implementing organizations [2, 7, 10, 22].

These concerns are necessary, but they do not fully address how systems define success *before* deployment. A model may be transparent, auditable, and technically accurate while still optimizing toward a narrow institutional goal. In humanitarian contexts, common goals include speed, coverage, predictive accuracy, cost reduction, logistical efficiency, and rapid delivery. These objectives may be reasonable under crisis conditions, but they are not neutral. They determine what the system prioritizes, what tradeoffs it accepts, and which forms of need become actionable. The governance question is therefore not only whether an AI system performs well, but what it has been asked to optimize [11, 17].

A smaller body of work has begun to connect optimization choices with governance concerns. This work shows that metrics, target variables, proxies, and objective functions translate broad values into operational rules [17]. In fairness and optimization research, different mathematical formulations can represent different theories of equity [19, 23, 24]. A system designed to maximize total welfare may behave differently from one designed to prioritize the worst-off group. Similarly, a system that balances fairness and utility requires human judgment about which tradeoffs are acceptable. These choices cannot be resolved through technical performance alone because they reflect normative judgments about whose needs matter and how competing goals should be balanced [18, 19].

This problem is especially visible in humanitarian assistance allocation. Aid distribution requires decisions about who should receive scarce resources, how quickly aid should be delivered, and whether priority should be given to reaching the largest number of people or to assisting those identified as most vulnerable. Prior work on food distribution shows that changing the weight given to equity, effectiveness, and efficiency can produce different allocation policies [20]. Work on refugee camp supply systems similarly shows that technical design choices require co-design and accountability because they shape how aid systems classify, monitor, and prioritize affected populations [22]. These differences are not simply implementation details. They reflect the priorities embedded in the allocation rule.

In this sense, the objective function operates as a governance mechanism. It helps define the unit of analysis, identify what counts as “need,” and determine how “success” is measured. If a system is optimized for proximity, it may reach more households quickly but miss distant household with significantly higher need. If it is optimized for vulnerability, it may concentrate resources among a smaller set of prioritized households, trading off the larger number of households that could have received assistance. Neither objective is inherently correct. Each reflects a different

understanding of humanitarian priority and produces different distributional consequences.

Despite a growing recognition of this tradeoff, the governance implications of objective-function selection are rarely examined directly. Work on humanitarian AI governance has largely concentrated on outcomes and processes [11], and although auditing has been proposed as a way to hold AI systems to account [13], audits and reviews typically examine deployed systems and their outputs rather than the objective a system was set to pursue. Formal optimization research, for its part, has produced ways of representing and balancing competing objectives, from social welfare functions to fairness–utility formulations that make value tradeoffs explicit and adjustable [19, 20]. Related work in humanitarian operations has shown that changing the weights or design choices in an allocation model can change the resulting policy [25]. Yet these choices are not arranged to isolate the effect of the objective itself. Governance and auditing look mainly downstream of choices, while optimization generally focuses on a chosen objective rather than treating the choice between objectives as the variable of interest.

What is missing is a controlled setting in which to vary the allocation objective on its own, while the population, the available resources, and the operating conditions are held constant to observe the distributional consequences of that single choice. Auditing real-world deployed systems cannot easily provide this, because real operations do not serve the same population under several competing objectives, and because affected people cannot ethically be subjected to such experiments. Simulation offers one way to address this gap. Agent-based modeling specifically allows us to represent heterogeneous households and aid agencies explicitly, vary the aid allocation, and hold everything else fixed [18, 26]. Here, we explain the agent-based model representing simple aid-distribution rules, described in the next section, to examine how throughput-based and vulnerability-based objectives shape the distribution of assistance and unmet need.

3 Model

3.1 Model Description

We built an agent-based model (using *NetLogo*) to test the theoretical distribution of aid along several different logics proposed in our previous work [11]. The model represents a humanitarian aid distribution system consisting of households and aid agencies. Households are spatially distributed across the environment and possess heterogeneous levels of need and vulnerability. Aid agencies distribute a finite stock of aid packages to households according to a selected allocation strategy.

Households are characterized by five attributes: need, distance to the nearest aid agency, two dimensions of vulnerability (Vulnerability-A and Vulnerability-B), and the number of aid packages received. We draw the two vulnerability measures from separate gamma distributions to represent different vulnerability profiles within the population. The model also calculates a composite vulnerability score (Vulnerability-

C) as the average of the two dimensions. Initial household need is generated as a weighted combination of vulnerability and a stochastic term drawn from a gamma distribution. Aid agencies are characterized by their spatial location, available aid inventory, and the number of aid packages distributed during the simulation.

To explore how different governance choices shape humanitarian outcomes, we implement two allocation strategies. Under *throughput optimization*, agencies allocate aid to the nearest households with unmet need. Under *vulnerability optimization*, agencies allocate aid based on household vulnerability rather than proximity. We examine three vulnerability formulations: Vulnerability-A, Vulnerability-B, and a composite measure (Vulnerability-C) that averages the two dimensions. To approximate a triage strategy of distributing available aid to the most vulnerable households, the model first identifies households with the highest vulnerability dimension and high need, then selects the top quartile of those to be “eligible households” for aid distribution, and then prioritizes aid distribution within that subset.

Each simulation begins by randomly placing households and aid agencies within the environment. Agencies receive an initial stock of aid packages, while households receive heterogeneous *need* and *vulnerability* values. During each time step, agencies distribute aid according to the selected allocation strategy. Aid reduces household need according to an aid-effectiveness parameter, and households whose need falls below a threshold exit the eligible recipient pool. The simulation terminates when agencies exhaust available aid or when no households remain in need of assistance.

These allocation strategies reflect different governance logics. Throughput optimization prioritizes efficient delivery to the most households, while vulnerability optimization prioritizes households identified as being most vulnerable along different (theoretical) dimensions of vulnerability (a, b, and c). The model therefore allows comparison of how alternative definitions of priority influence humanitarian outcomes under otherwise similar operating conditions.

3.2 Sensitivity Analysis

We used *BehaviorSearch* to examine the sensitivity of model outcomes to four parameters [27]: the number of households, the number of aid agencies, aid inventory, and aid effectiveness. For each allocation strategy, we identified parameter combinations that maximized and minimized average household need over the simulation run.

Table 1. Summary of Sensitivity Analysis Findings.

<i>Parameter</i>	<i>Influence on Outcomes</i>
<i>Aid impact</i>	Strong
<i>Number of agencies</i>	Moderate; greater influence under throughput allocation
<i>Aid inventory</i>	Limited
<i>Vulnerability formulation (A, B, C)</i>	Limited differences in aggregate outcomes

The analysis revealed three broad patterns. First, aid impact was the strongest driver of model performance. Second, throughput and vulnerability-based allocation responded differently to changes in model parameters, particularly those related to geography and agency distribution. Third, Vulnerability-A, Vulnerability-B, and the composite vulnerability measure produced similar aggregate outcomes across the tested parameter ranges as we would expect given they share distribution type (gamma) with varying parameter values (see **Table 1**).

Overall, the sensitivity analysis suggested that model behavior was influenced more by the choice of allocation objective than by the specific formulation of vulnerability. The behavior of the model aligned with our theoretical design choices about aid allocation.

4 Results

We compared throughput optimization with three vulnerability-based allocation strategies (Vulnerability-A, Vulnerability-B, and Composite Vulnerability) under varying levels of resource scarcity. Throughput allocation prioritized proximity to aid agencies, while vulnerability-based approaches prioritized households with the highest vulnerability scores.

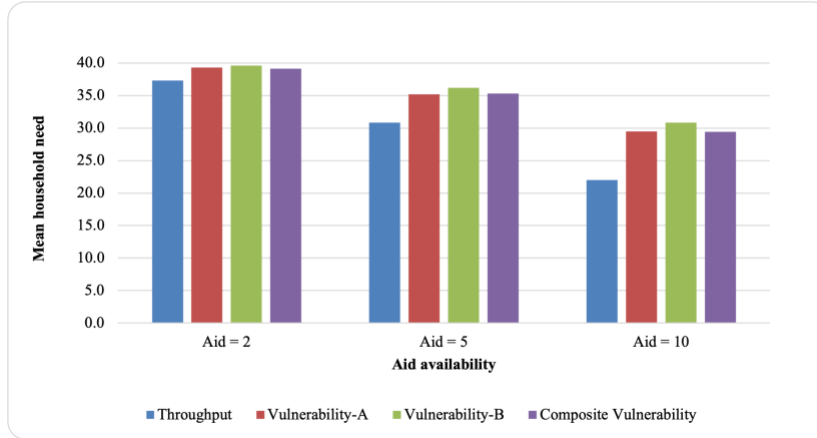


Figure 1. Mean Household Need by Allocation Strategy and Aid Availability.

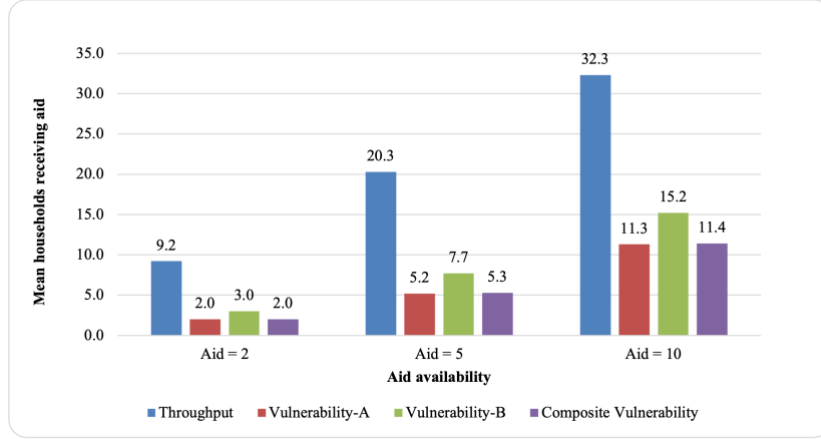


Figure 2. Mean number of households receiving aid by allocation strategy and aid availability.

As expected, increasing aid availability reduced average household need under all allocation strategies. More interestingly, the choice of the objective function also shaped outcomes. Across all aid conditions, throughput optimization produced a lower average need than any of the vulnerability-based approaches. Under the highest aid condition, throughput reduced the average need to approximately 22 units, compared to 29–31 units under the vulnerability-based strategies (see **Figure 1**). Values represent the mean household need at the end of the simulation, averaged across model runs.

Throughput also reached substantially more households than any of the vulnerability-based approaches, as we would expect (see **Figure 2**). Under the highest aid condition, throughput reached roughly one-third of households, while vulnerability-based allocation concentrated assistance among a much smaller subset of recipients. This difference follows directly from the objectives embedded in each allocation strategy. Throughput attempts to reduce the need across as many households as possible. Vulnerability-based approaches direct aid toward households identified as most vulnerable, even if that means serving fewer households overall.

4.1 Implications for Humanitarian AI Governance

The simulation results reinforce a central argument from our earlier work that optimization objectives are not merely technical design decisions but governance choices [11]. In our previous humanitarian AI audit framework paper [11], we argued that humanitarian AI systems frequently optimize for speed, coverage, predictive accuracy, or logistical efficiency, while treating questions of equity and intersectional vulnerability as secondary considerations addressed after deployment. The model provides a simple demonstration of this dynamic.

When the system was optimized for throughput, it reduced population-wide need more effectively and reached a larger share of households. From an efficiency

perspective, this would appear to be the preferred outcome. However, vulnerability-based allocation embodied a different objective of prioritizing households identified as most vulnerable rather than maximizing aggregate need reduction. The resulting tradeoff mirrors the efficiency-equity tension identified in humanitarian AI systems such as disaster assessment platforms, anticipatory action systems, and aid targeting algorithms.

Table 2. Governance Tradeoffs Observed Across Allocation Objectives

<i>Allocation Strategy</i>	<i>Primary Optimization Logic</i>	<i>Population-Level Outcome</i>	<i>Distributional Outcome</i>
<i>Throughput</i>	Geographic proximity	Lowest average household need across all aid conditions	Aid distributed across a larger share of households
<i>Vulnerability-A</i>	Prioritize highest Vulnerability-A households	Higher average need than throughput	Aid concentrated among highly vulnerable households
<i>Vulnerability-B</i>	Prioritize highest Vulnerability-B households	Similar aggregate outcomes to Vulnerability-A	Reached slightly more households than other vulnerability measures
<i>Composite Vulnerability</i>	Balance multiple vulnerability dimensions	Similar aggregate outcomes to Vulnerability-A	Allocation concentrated among households with high composite vulnerability

Importantly, the model does not suggest that one objective is inherently superior. Rather, it illustrates how different optimization targets produce different distributions of benefits and burdens. Aggregate measures such as average need favor throughput-oriented approaches, while vulnerability-oriented approaches direct resources toward a smaller subset of prioritized households (see **Table 2**). While it may seem effective to reach the largest number of households, equity design choices would ensure that those most in need would receive the most aid. In this sense, the objective function itself becomes a governance mechanism that determines whose needs count, whose vulnerabilities are made actionable, and what constitutes success.

5 Conclusion

The simulation results provide a simple illustration of a broader governance concern raised in humanitarian AI. Across all experiments, changing the allocation objective altered both who received assistance and the resulting distribution of unmet need. Throughput optimization reduced average need across the population and reached a larger number of households, while vulnerability-based approaches concentrated resources among a smaller set of prioritized recipients. These results followed directly from the objective function the system was instructed to optimize.

This distinction echoes a central argument from our earlier work on humanitarian AI governance. AI systems do not independently determine what constitutes a desirable outcome; they optimize the objectives, constraints, and units of analysis specified by designers and institutions. Once selected, these objectives become embedded within operational decision-making and shape whose needs are made visible, actionable, and ultimately prioritized. The question is therefore not simply whether an AI system performs accurately, but what it has been asked to optimize in the first place.

While the model presented here is intentionally simplified, it demonstrates how different optimization objectives can produce materially different allocation outcomes even when operating on the same population and resource base. In this sense, objective functions should be understood not merely as technical parameters but as governance choices with distributive consequences, particularly when the stakes are high in humanitarian response attempting to balance intersectional needs and vulnerabilities.

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References

1. Nearing, G., et al., *Global prediction of extreme floods in ungauged watersheds*. Nature, 2024. **627**(8004): p. 559–563.
2. Comes, T., *AI for crisis decisions*. Ethics and Information Technology, 2024. **26**(1): p. 12.
3. Veber, M.T., *International Organizations and AI-Supported Humanitarian Aid: Navigating through the Applicable (Data Protection) Legal Regimes*. International and comparative law review (Olomouc, Czech Republic), 2024. **24**(2): p. 54–83.
4. Van den Homberg, M.J.C., C.M. Gevaert, and Y. Georgiadou, *The Changing Face of Accountability in Humanitarianism: Using Artificial Intelligence for Anticipatory Action*. Politics and governance, 2020. **8**(4): p. 456–467.
5. Lythreath, S., F. Acikgoz, and N. Yassine, *Artificial intelligence in humanitarian aid: A review and future research agenda*. Technovation, 2026. **151**: p. 103415.
6. Ramiah, D., *5 ways AI can help crisis response around the world*. 2024: UNDP.
7. Beduschi, A., *Harnessing the potential of artificial intelligence for humanitarian action: Opportunities and risks*. International Review of the Red Cross, 2022. **104**(919): p. 1149–1169.
8. Haykal, D., Goldust, M., Cartier, H., & Treacy, P., *AI in humanitarian healthcare: a game changer for crisis response*. Frontiers in artificial intelligence, 2025. **8**: p. 1627773. <https://doi.org/10.3389/frai.2025.1627773>.
9. Dubey, R., et al., *Impact of artificial intelligence-driven big data analytics culture on agility and resilience in humanitarian supply chain: A practice-based view*. International Journal of Production Economics, 2022. **250**: p. 108618.

10. Gevaert, C.M., et al., *Fairness and accountability of AI in disaster risk management: Opportunities and challenges*. Patterns, 2021. **2**(11).
11. Huseynli, G., Erika Frydenlund. *The efficiency–equity tradeoff in AI-enabled humanitarian information systems: Toward an intersectional governance audit framework*. in *23rd International Conference on Information Systems for Crisis Response and Management (ISCRAM)*. 2026. The Hague, Netherlands: (in press).
12. Madianou, M., *Nonhuman humanitarianism: when 'AI for good' can be harmful*. Information, Communication & Society, 2021. **24**(6): p. 850–868.
13. Sandu, I., M. Wiersma, and D. Manichand, *Time to audit your AI algorithms*. MAB ('s-Gravenhage. Online), 2022. **96**(7/8): p. 253–265.
14. Noble, S.U., *Algorithms of Oppression: How Search Engines Reinforce Racism*. 2018, New York, USA: New York University Press.
15. Crawford, K., *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. 2021: Yale University Press.
16. Molnar, P., *The walls have eyes: Surviving migration in the age of artificial intelligence*. 2024: The New Press.
17. Kuziemski, M. and G. Misuraca, *AI governance in the public sector: Three tales from the frontiers of automated decision-making in democratic settings*. Telecommunications Policy, 2020. **44**(6): p. 101976.
18. Xinying Chen, V. and J.N. Hooker, *A guide to formulating fairness in an optimization model*. Annals of Operations Research, 2023. **326**(1): p. 581–619.
19. Zhang, Y., R. Bellamy, and K. Varshney, *Joint Optimization of AI Fairness and Utility: A Human-Centered Approach*, in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 2020, Association for Computing Machinery: New York, NY, USA. p. 400–406.
20. Hasnain, T., I. Sengul Orgut, and J.S. Ivy, *Elicitation of Preference among Multiple Criteria in Food Distribution by Food Banks*. Production and Operations Management, 2021. **30**(12): p. 4475–4500.
21. UNHCR, *UNHCR Blogs*. 2025.
22. Dekker, R., et al., *Co-designing algorithms for governance: Ensuring responsible and accountable algorithmic management of refugee camp supplies*. Big Data & Society, 2022. **9**(1): p. 20539517221087855.
23. Robertson, J., et al. *A human-in-the-loop fairness-aware model selection framework for complex fairness objective landscapes*. in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 2024.
24. Kubes, T., *A Critique of Human-Centred AI: A Plea for a Feminist AI Framework (FAIF)*. AI & society, 2025.
25. Gupta, S., et al., *Too Many Fairness Metrics: Is There a Solution? Equity Across Demographic Groups for the Facility Location Problem*, in *Data Science and Optimization*. 2020, Springer. p. 231–272.
26. Cohen, T. and N.P. Suzor, *Contesting the public interest in AI governance*. Internet policy review, 2024. **13**(3).
27. Miller, J.H., *Active nonlinear tests (ANTs) of complex simulation models*. Management Science, 1998. **44**(6): p. 820–830.